

Nucleophilic substitution at the carbonyl group

10

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Arriving at

- Nucleophilic attack followed by loss of leaving group
- What makes a good nucleophile
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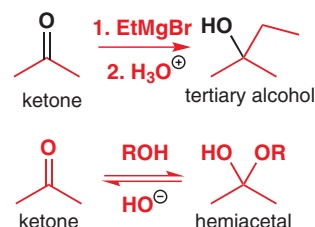
You are already familiar with reactions of compounds containing carbonyl groups. Aldehydes and ketones react with nucleophiles at the carbon atom of their carbonyl group to give products containing hydroxyl groups. Because the carbonyl group is such a good electrophile, it reacts with a wide range of different nucleophiles: you have met reactions of aldehydes and ketones with (in Chapter 6) cyanide, water, and alcohols, and (in Chapter 9) organometallic reagents (organolithiums and organomagnesiums, or Grignard reagents).

In this chapter and Chapter 11 we shall look at some more reactions of the carbonyl group—and revisit some of the ones we touched on in Chapter 6. It is a tribute to the importance of this functional group for organic chemistry that we have devoted four chapters of this book to its reactions. Just like the reactions in Chapters 6 and 9, the reactions in Chapters 10 and 11 all involve attack of a nucleophile on a carbonyl group. The difference is that this step is followed by other mechanistic steps, which means that the overall reactions are not just *additions* but also *substitutions*.

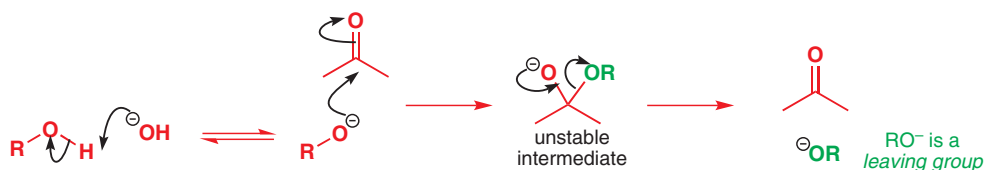
The product of nucleophilic addition to a carbonyl group is not always a stable compound

Addition of a Grignard reagent to an aldehyde or ketone gives a stable alkoxide, which can be protonated with acid to produce an alcohol (you met this reaction in Chapter 9). The same is not true for addition of an alcohol to a carbonyl group in the presence of base—in Chapter 6 we drew a reversible, equilibrium arrow for this transformation and said that the product, a hemiacetal, is formed to a significant extent only if it is cyclic.

The reason for this instability is that RO^- is easily expelled from the molecule. We call groups that can be expelled from molecules, usually taking with them a negative charge, **leaving groups**. We'll look at leaving groups in more detail later in this chapter and again in Chapter 15.



Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

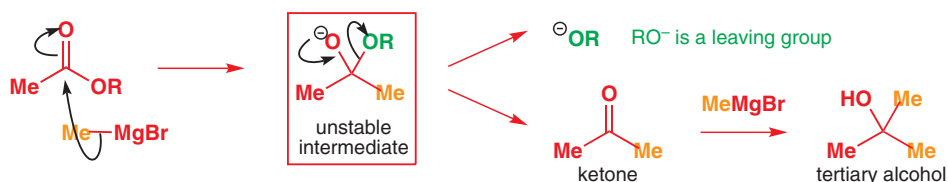


Leaving groups

Leaving groups are anions such as Cl^- , RO^- , and RCO_2^- that can be expelled from molecules taking their negative charge with them.

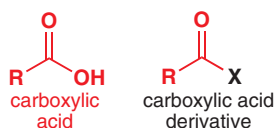
So, if the nucleophile is also a leaving group, there is a chance that it will be lost again and that the carbonyl group will reform—in other words, the reaction will be reversible. The energy released in forming the $\text{C}=\text{O}$ bond (bond strength 720 kJ mol^{-1}) makes up for the loss of two $\text{C}-\text{O}$ single bonds (about 350 kJ mol^{-1} each), one of the reasons for the instability of the hemiacetal product in this case.

The same thing can happen if the starting carbonyl compound contains a potential leaving group. The unstable negatively charged intermediate in the red box below is formed when a Grignard reagent is added to an ester.



Again, it collapses with loss of RO^- as a leaving group. This time, though, we have not gone back to starting materials: instead we have made a new compound (a ketone) by a **substitution reaction**—the OR group of the starting material has been substituted by the Me group of the product. In fact the ketone product can react with the Grignard reagent a second time to give a tertiary alcohol. Later in this chapter we'll discuss why the reaction doesn't stop at the ketone.

Carboxylic acid derivatives



Most of the starting materials for, and products of, these substitutions will be carboxylic acid derivatives, with the general formula RCOX . You met the most important members of this class in Chapter 2: here they are again as a reminder.

Carboxylic acid derivatives

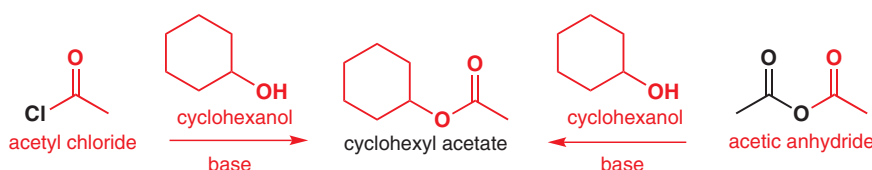
Carboxylic acid	Derivative of RCO_2H			
	acid chloride or acyl chloride*	ester	acid anhydride	amide

*We shall use these two terms interchangeably.

■ The reactions of alcohols with acid chlorides and with acid anhydrides are the most important ways of making esters, but not the only ways. We shall see later how carboxylic acids can be made to react directly with alcohols.

Acid chlorides and acid anhydrides react with alcohols to make esters

Acetyl chloride will react with an alcohol in the presence of a base to give an acetate ester and we get the same product if we use acetic anhydride.

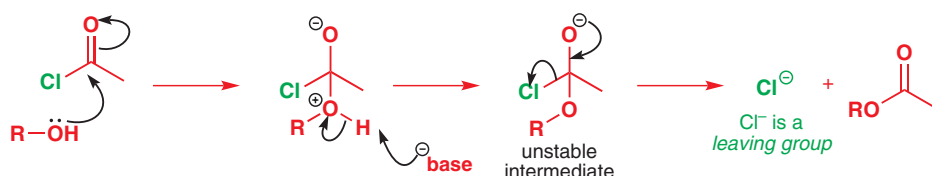


Remember the symbol for acetyl? $\text{Ac} = \text{CH}_3\text{CO}$. You can represent the acetate of an alcohol ROH as ROAc but not as RAc as this would be a ketone.

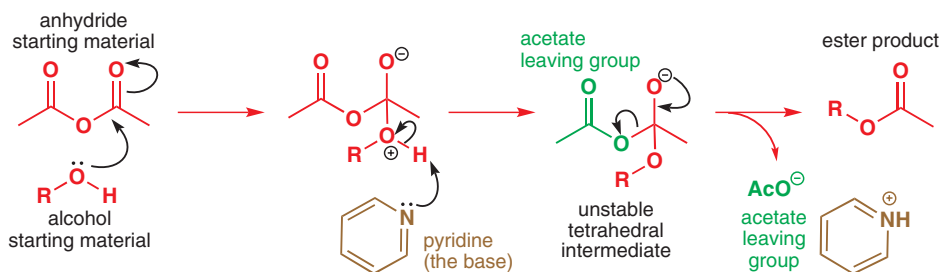
In each case, a substitution (of the black part of the molecule, Cl^- or AcO^- , by cyclohexanol) has taken place—but how? It is important that you learn not only the *fact* that acyl chlorides and acid anhydrides react with alcohols but also the *mechanism* of the reaction. In this chapter you will meet a lot of reactions, but relatively few mechanisms—once you understand one, you should find that the rest follow on quite logically.

The first step of the reaction is, as you might expect, addition of the nucleophilic alcohol to the electrophilic carbonyl group—we'll take the acyl chloride first. The base is important because it removes the proton from the alcohol once it attacks the carbonyl group. A base commonly used for this is pyridine. If the electrophile had been an aldehyde or a ketone, we would have got an unstable hemiacetal, which would collapse back to starting materials by eliminating the alcohol. With an acyl chloride, the alkoxide intermediate we get is also unstable. It collapses again by an elimination reaction, this time losing chloride ion, to form the ester. Chloride is the *leaving group* here—it leaves with its negative charge.

You will notice that the terms 'acid chloride' and 'acyl chloride' are used interchangeably.



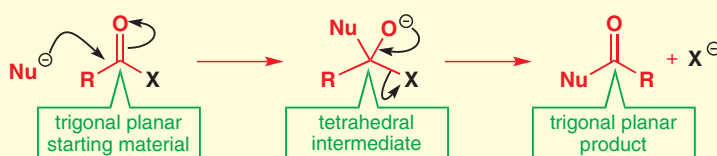
With this reaction as a model, you should be able to work out the mechanism of ester formation from acetic anhydride and an alcohol. Try to write it down without looking at the acyl chloride mechanism above, and certainly not at the answer below. Here it is, with pyridine as the base. Again, addition of the nucleophile gives an unstable intermediate, which undergoes an elimination reaction, this time losing a carboxylate anion to give an ester.



We call the unstable intermediate formed in these reactions the **tetrahedral intermediate** because the trigonal (sp^2) carbon atom of the carbonyl group has become a tetrahedral (sp^3) carbon atom.

● Tetrahedral intermediates

Substitutions at trigonal carbonyl groups go through a tetrahedral intermediate and then on to a trigonal product.



More details of this reaction

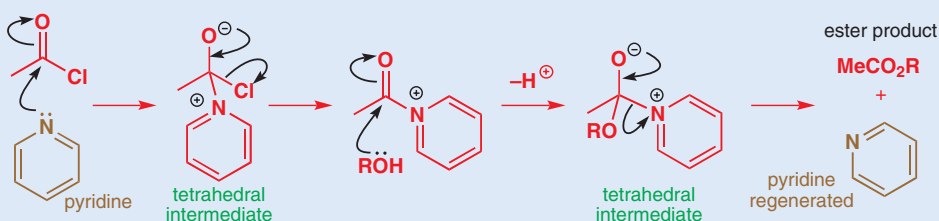
Acylation with acyl chlorides in the presence of pyridine has more subtleties than first meet the eye. If you are reading this chapter for the first time, you might skip this box, as it is not essential to the general flow of what we are saying. There are three more points to notice.

Pyridine is consumed during both of these reactions, since it ends up protonated. One whole equivalent of pyridine is therefore necessary and, in fact, the reactions are often carried out with pyridine as solvent.

The observant among you may also have noticed that the (weak—pyridine) base catalyst in this reaction works very slightly differently from the (strong—hydroxide) base catalyst in the hemiacetal-forming reaction on p. 197: pyridine removes the proton *after* the nucleophile has added; hydroxide removes the proton *before* the nucleophile has added. This is deliberate, and will be discussed further in Chapters 12 and 40. The basicities of pyridine (pK_a for protonation 5.5) and hydroxide (pK_a of water 15.7) were discussed in Chapter 8.

Pyridine is, in fact, more nucleophilic than the alcohol, and it attacks the acyl chloride rapidly, forming a highly electrophilic (because of the positive charge) intermediate. It is then this intermediate that subsequently reacts with the alcohol to give the ester. Because pyridine is acting as a nucleophile to speed up the reaction, yet is unchanged by the reaction, it is called a **nucleophilic catalyst**.

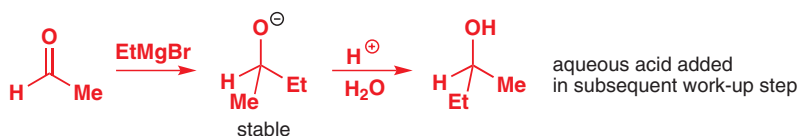
nucleophilic catalysis in ester formation



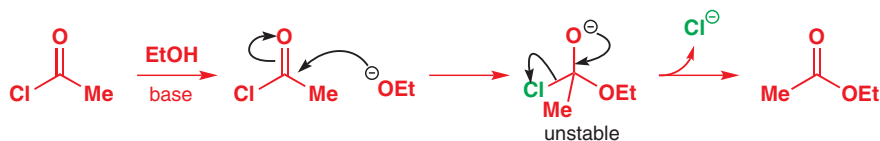
Interactive mechanism for pyridine nucleophilic catalysis

Why are the tetrahedral intermediates unstable?

The alkoxide formed by addition of a Grignard reagent to an aldehyde or ketone is stable, lasting long enough to be protonated on work-up in acid to give an alcohol as product.



Tetrahedral intermediates are similarly formed by addition of a nucleophile, say ethanol in base, to the carbonyl group of acetyl chloride, but these tetrahedral intermediates are unstable. Why are they *unstable*? The answer is to do with leaving group ability. Once the nucleophile has added to the carbonyl compound, the stability of the product (or tetrahedral intermediate) depends on how good the groups attached to the new tetrahedral carbon atom are at leaving with the negative charge. In order for the tetrahedral intermediate to collapse (and therefore be just an intermediate and not the final product) one of the groups has to be able to leave and carry off the negative charge from the alkoxide anion formed in the addition.



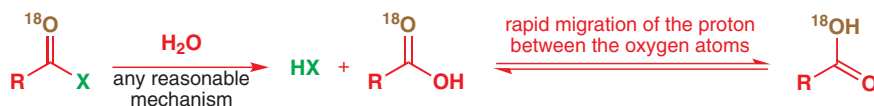
The most stable anion will be the best leaving group. There were three choices for the leaving group: Cl^- , EtO^- , or Me^- . We can make MeLi but not Me^- because it is very unstable so Me^- must be a very bad leaving group. EtO^- is not so bad—alkoxide salts are stable, but they are still strong, reactive bases. But Cl^- is the best leaving group: Cl^- ions are perfectly stable and quite unreactive, and happily carry off the negative charge from the oxygen atom.

You probably eat several grams of Cl^- every day but you would be unwise to eat EtO^- or MeLi . So neither of these reactions occurs:



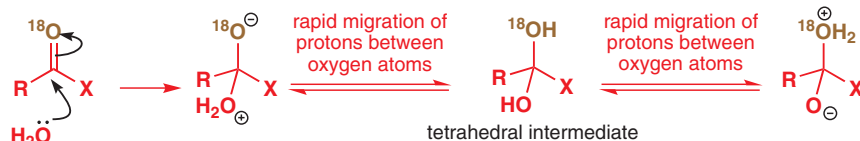
How do we know that the tetrahedral intermediate exists?

We don't expect you to be satisfied with the bland statement that tetrahedral intermediates are formed in these reactions: of course, you wonder how we know that this is true. The first evidence for tetrahedral intermediates in the substitution reactions of carboxylic acid derivatives was provided by Bender in 1951. He made carboxylic acid derivatives RCOX that had been 'labelled' with an isotope of oxygen, ^{18}O . This is a non-radioactive isotope that is detected by mass spectrometry. He then reacted these derivatives with water to make labelled carboxylic acids. By any reasonable mechanism, the products would have one ^{18}O atom from the labelled starting material. Because the proton on a carboxylic acid migrates rapidly from one oxygen to another, both oxygens are labelled equally.

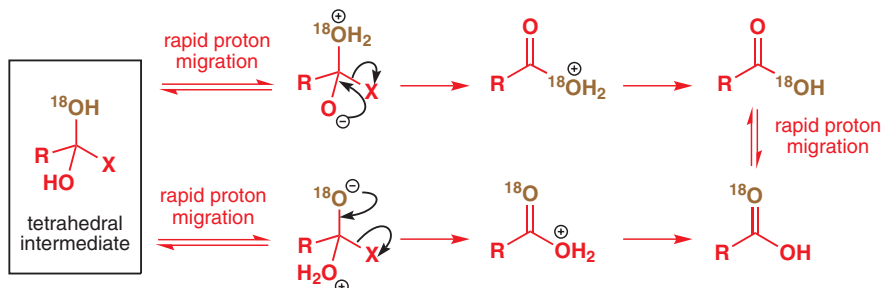


■ In Bender's original work, X was an alkoxy group (i.e. RCOX was an ester).

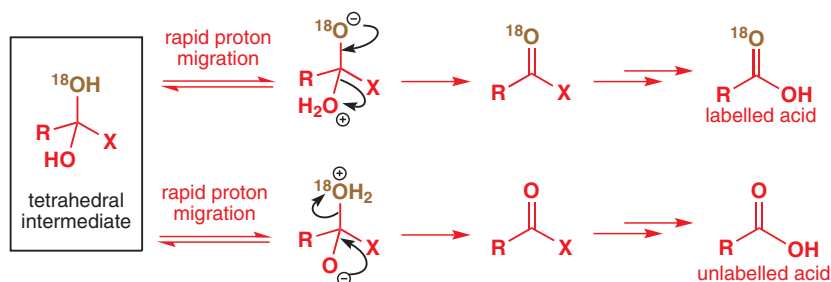
He then reacted these derivatives with insufficient water for complete consumption of the starting material. At the end of the reaction, he found that the proportion of labelled molecules in the *remaining starting material* had decreased significantly: in other words, it was no longer completely labelled with ^{18}O ; some contained 'normal' ^{16}O . The formation of the tetrahedral intermediate would be as before but rapid proton transfer would also mean that the two oxygen atoms would be the same. Now you may see the next step in the argument.



This result cannot be explained by direct substitution of X by H_2O , but is consistent with the existence of an intermediate in which the unlabelled ^{16}O and labelled ^{18}O can 'change places'. This intermediate is the *tetrahedral intermediate* for this reaction. Either isomer can lose X and in each case labelled carboxylic acid is formed.



But either tetrahedral intermediate could lose water instead. In one case (top line below) the original starting material is regenerated complete with label. But in the second case, labelled water is lost and *unlabelled starting material* is formed. This result would be difficult to explain without a tetrahedral intermediate with a lifetime long enough to allow for proton exchange. This 'addition-elimination' mechanism is now universally accepted.



pK_a is a useful guide to leaving group ability

It's useful to be able to compare leaving group ability quantitatively. This is impossible to do exactly, but a good guide is the pK_a of the conjugate acid (Chapter 8). If X^- is the leaving group, the lower the pK_a of HX, the better X^- is as a leaving group. If we go back to the example of ester formation from acyl chloride plus alcohol, there's a choice of Me^- , EtO^- , and Cl^- . HCl is a stronger acid than EtOH, which is a much stronger acid than methane. So Cl^- is the best leaving group and EtO^- the next best. These observations apply only to reactions at the carbonyl group.

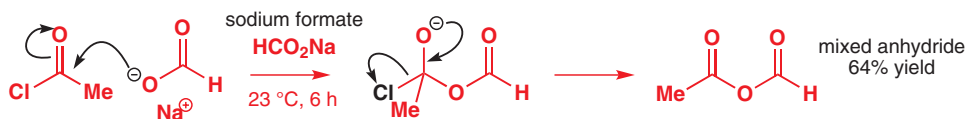
• Leaving group ability

The lower the pK_a of HX, the better the leaving group of X^- in carbonyl substitution reactions.

The most important substituents in carbonyl reactions are alkyl or aryl groups (R), amino groups in amides (NH_2), alkoxy groups in esters (RO^-), carboxylate groups (RCO_2^-) in anhydrides, and chloride (Cl^-) in acyl chlorides. The order of leaving group ability is then:

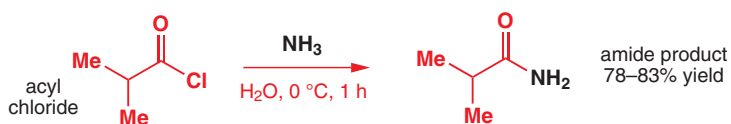
carboxylic acid derivative	leaving group, X^-	conjugate acid, HX	pK_a of HX	leaving group?
acyl chloride	Cl^-	HCl	<0	excellent
anhydride	RCOO^-	RCO_2H	about 5	good
ester	RO^-	ROH	about 15	poor
amide	NH_2^-	NH_3	about 25	very poor
alkyl or aryl derivative	R^-	RH	>40	not a leaving group

We can use pK_a to predict what happens if we react an acyl chloride with a carboxylate salt. We expect the carboxylate salt (here, sodium formate or sodium methanoate, HCO_2Na) to act as the nucleophile to form a tetrahedral intermediate, which could collapse in any one of three ways. We can straightaway rule out loss of Me^- and we might guess that Cl^- is a better leaving group than HCO_2^- as HCl is a much stronger acid than a carboxylic acid, and we'd be right. Sodium formate reacts with acetyl chloride to give a mixed anhydride.

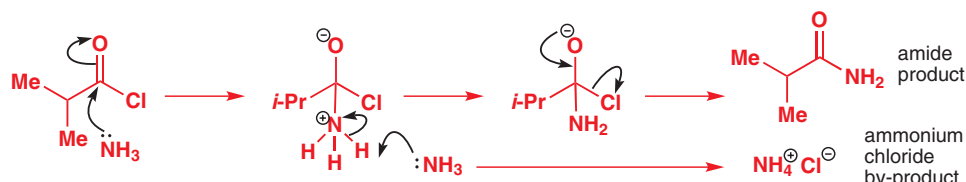


Amines react with acyl chlorides to give amides

Using the principles we've outlined above, you should be able to see how these compounds can be interconverted by substitution reactions with appropriate nucleophiles. We've seen that acid chlorides react with carboxylic acids to give acid anhydrides, and with alcohols to give esters. They also react with amines (such as ammonia) to give amides.

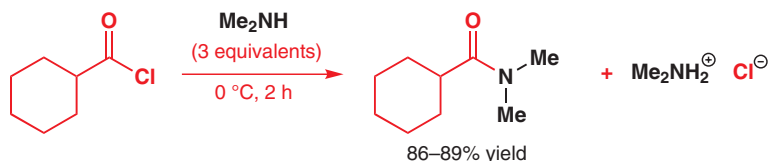


The mechanism is very similar to the mechanism of ester formation. Notice the second molecule of ammonia, which removes a proton, and the loss of chloride ion—the leaving group—to form the amide. Ammonium chloride is formed as a by-product in the reaction.



Interactive mechanism for amide formation

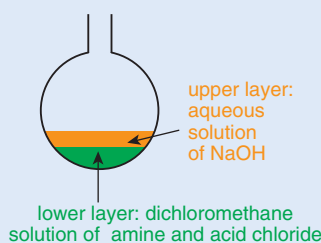
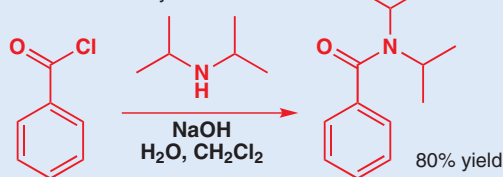
Here is another example, using a secondary amine, dimethylamine. Try writing down the mechanism now without looking at the one above. Again, two equivalents of dimethylamine are necessary, although the chemists who published this reaction added three for good measure.



Schotten–Baumann synthesis of an amide

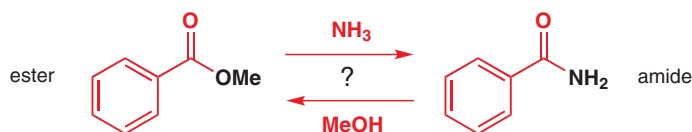
As these mechanisms show, the formation of amides from acid chlorides and amines is accompanied by production of one equivalent of HCl , which needs to be neutralized by a second equivalent of amine. An alternative method for making amides is to carry out the reaction in the presence of another base, such as NaOH , which then does the job of neutralizing the HCl . The trouble is, OH^- also attacks acyl chlorides to give carboxylic acids. Schotten and Baumann, in the late nineteenth century, published a way round this problem by carrying out these reactions in two-phase systems of immiscible water and dichloromethane. The organic amine (not necessarily ammonia) and the acyl chloride remain in the (lower) dichloromethane layer, while the base (NaOH) remains in the (upper) aqueous layer. Dichloromethane and chloroform are two common organic solvents that are heavier (more dense) than water. The acyl chloride reacts only with the amine, but the HCl produced can dissolve in, and be neutralized by, the aqueous solution of NaOH .

Schotten–Baumann synthesis of an amide

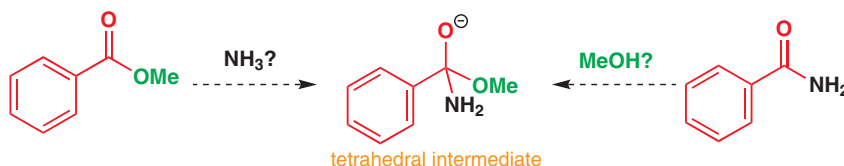


Using base strength to predict the outcome of substitution reactions of carboxylic acid derivatives

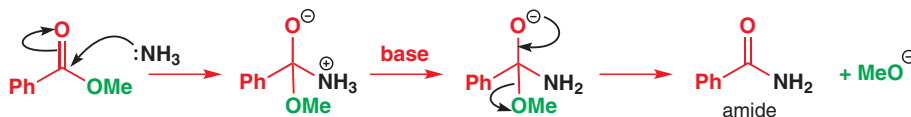
You saw that acid anhydrides react with alcohols to give esters; they will also react with amines to give amides. But would you expect esters to react with amines to give amides, or amides to react with alcohols to give esters? Both appear reasonable.



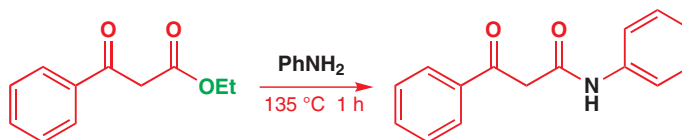
In fact only the top reaction works: amides can be formed from esters but esters cannot be formed from amides. The key question is: which group will leave from the common tetrahedral intermediate? The answer is MeO^- and not NH_2^- . You should have worked this out from the stability of the anions. Alkoxides are reasonably strong bases ($\text{p}K_{\text{a}}$ of ROH about 15) so they are not good leaving groups. But NH_2^- is a very unstable anion ($\text{p}K_{\text{a}}$ of NH_3 about 25) and is a very bad leaving group.



So MeO^- leaves and the amide is formed. The base used to deprotonate the first formed intermediate may be either the MeO^- produced in the reaction or, to start with, another molecule of NH_3 .



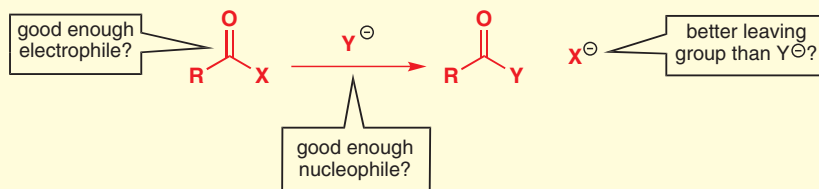
Here is a slightly unusual example in that there is a ketone present in the molecule as well. Later in the book we shall consider how to work out whether another functional group might interfere with the reaction we want to do.



Factors other than leaving group ability can be important

In fact, the tetrahedral intermediate would simply never form from an amide and an alcohol; the amide is too bad an electrophile and the alcohol not a good enough nucleophile. We've looked at leaving group ability: next we'll consider the strength of the nucleophile Y^- and then the strength of the electrophile RCOX .

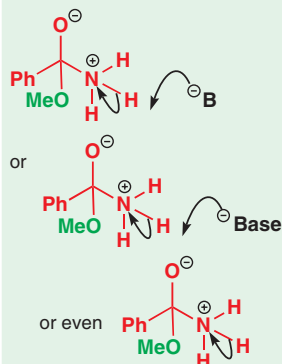
● Conditions for reaction



If this reaction is to go:

- 1 X^- must be a better leaving group than Y^- (otherwise the reverse reaction would take place).
- 2 Y^- must be a strong enough nucleophile to attack RCOX .
- 3 RCOX must be a good enough electrophile to react with Y^- .

■ You will meet many more mechanisms like this, in which an unspecified base removes a proton from an intermediate. As long as you can satisfy yourself that there is a base available to perform the task, it is quite acceptable to write any of these shorthand mechanisms.



Strength of nucleophile and leaving group ability are related and pK_a is a guide to both

We have seen how pK_a gives us a guide to leaving group ability: it is also a good guide to how strong a nucleophile will be. These two properties are the reverse of each other: good nucleophiles are bad leaving groups. A stable anion is a good leaving group but a poor nucleophile. Anions of weak acids (HA has high pK_a) are bad leaving groups but good nucleophiles towards the carbonyl group.

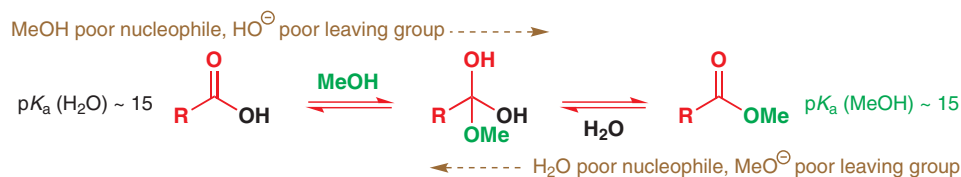
➔ We will come back to this concept again in Chapter 15, where you will see that this principle does not apply to substitution at saturated carbon atoms.

● Guide to nucleophilicity

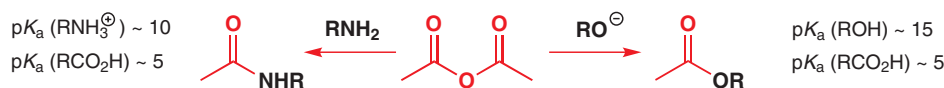
In general, the higher the pK_a of AH the better A^- is as a nucleophile.

But just a moment—we've overlooked an important point. We have sometimes used anions as nucleophiles (for example when we made acid anhydrides from acid chlorides plus carboxylate salts, we used an anionic nucleophile RCO_2^-) but on other occasions we have used neutral nucleophiles (for example when we made amides from acid chlorides plus amines, we used a neutral nucleophile NH_3). Anions are better nucleophiles for carbonyl groups than are neutral compounds so we can choose our nucleophilic reagent accordingly.

For proper comparisons, we should use the pK_a of NH_4^+ (about 10) if we are using neutral ammonia, but the pK_a of RCO_2H (about 5) if we're using the carboxylate anion. Ammonia is a good nucleophile and we don't usually need its anion but carboxylic acids are very weak nucleophiles and we often use their anions. You will see later in this chapter that we can alter this with acid catalysts. So this reaction works badly in either direction. We don't make or hydrolyse esters this way.

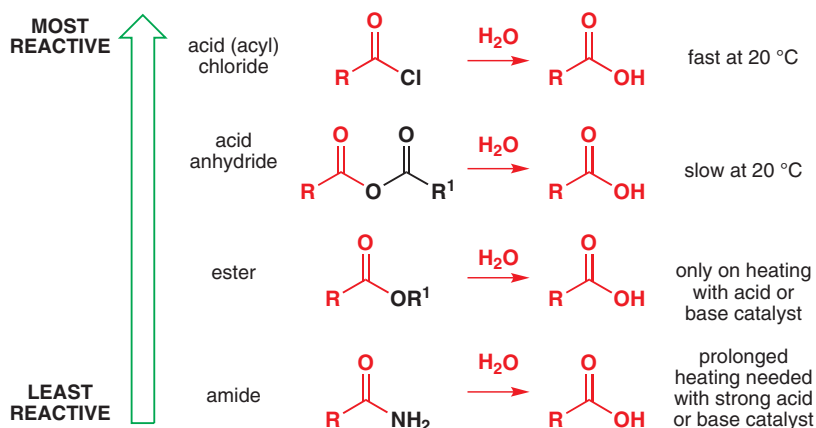


While amines react with acetic anhydride quite rapidly at room temperature (reaction complete in a few hours), alcohols react extremely slowly in the absence of a base. On the other hand, an alkoxide anion reacts with acetic anhydride extremely rapidly—the reactions are often complete within seconds at 0 °C. We don't have to deprotonate an alcohol completely to increase its reactivity: just a catalytic quantity of a weak base can do this job. All the pK_a s you need are in Chapter 8.



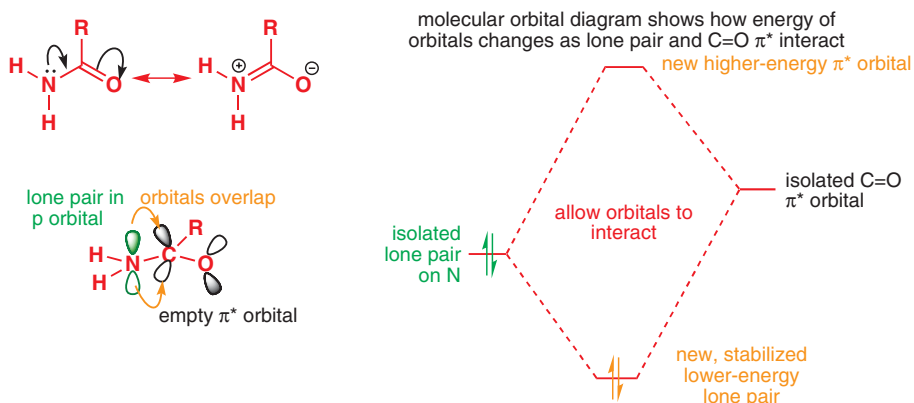
Not all carboxylic acid derivatives are equally reactive

We can list the common carboxylic acid derivatives in a 'hierarchy' of reactivity, with the most reactive at the top and the least reactive at the bottom. The nucleophile is the same in each case (water), as is the product, the carboxylic acid, but the electrophiles vary from very reactive to unreactive. The conditions needed for successful reaction show just how large is the variation on reactivity. Acid chlorides react violently with water. Amides need refluxing with 10% NaOH or concentrated HCl in a sealed tube at 100 °C overnight. We've seen that this hierarchy is partly due to how good the leaving group is (the ones at the top are best). But it also depends on the reactivity of the acid derivatives. Why is there such a large difference?

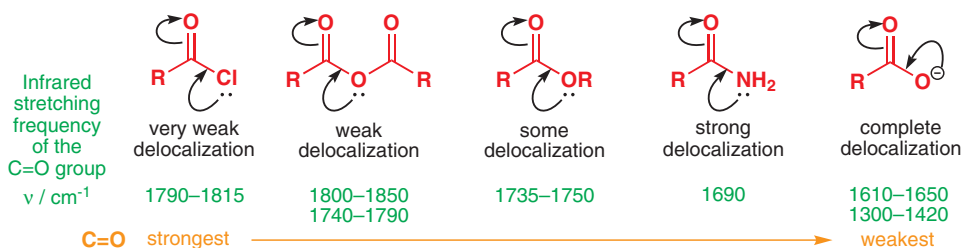


Delocalization and the electrophilicity of carbonyl compounds

Amides are the least reactive towards nucleophiles because they exhibit the greatest degree of delocalization. You met this concept in Chapter 7 and we shall return to it many times more. In an amide, the lone pair on the nitrogen atom can be stabilized by overlap with the π^* orbital of the carbonyl group—this overlap is best when the lone pair occupies a p orbital (in an amine, it would occupy an sp^3 orbital).



The molecular orbital diagram shows how this interaction both lowers the energy of the bonding orbital (the delocalized nitrogen lone pair), making it neither basic nor nucleophilic, and raises the energy of the π^* orbital, making it less ready to react with nucleophiles. Esters are similar, but because the oxygen lone pairs are lower in energy, the effect is less pronounced. The degree of delocalization depends on the electron-donating power of the substituent and increases along the series of compounds below from almost no delocalization from Cl to complete delocalization in the carboxylate anion, where the negative charge is equally shared between the two oxygen atoms.



The greater the degree of delocalization, the weaker the $\text{C}=\text{O}$ bond becomes. This is most clearly evident in the stretching frequency of the carbonyl group in the IR spectra of

carboxylic acid derivatives—remember that the stretching frequency depends on the force constant of the bond, itself a measure of the bond's strength. The carboxylate anion is included because it represents the limit of the series, with complete delocalization of the negative charge over the two oxygen atoms. There are two frequencies for the anhydride and the carboxylate anion because of symmetric and antisymmetric stretching of identical bonds.

Amides react as electrophiles only with powerful nucleophiles such as HO^- . Acid chlorides, on the other hand, react with even quite weak nucleophiles: neutral ROH , for example. They are more reactive because the electron-withdrawing effect of the chlorine atom increases the electrophilicity of the carbonyl carbon atom.

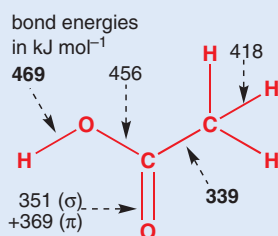
➔ Infrared spectroscopy was introduced in Chapter 3.

Bond strengths and reactivity

You may think that a weaker $\text{C}=\text{O}$ bond should be more reactive. This is not so because the partial positive charge on carbon is also lessened by delocalization and because the molecule as a whole is stabilized by the delocalization. Bond strength is not always a good guide to reactivity!

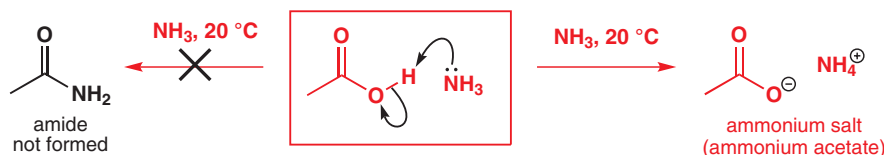
For example, in acetic acid the bond strengths are surprising. The strongest bond is the $\text{O}-\text{H}$ bond and the weakest is the $\text{C}-\text{C}$ bond. Yet very few reactions of acetic acid involve breaking the $\text{C}-\text{C}$ bond, and its characteristic reactivity, as an acid, involves breaking $\text{O}-\text{H}$, the strongest bond of them all!

The reason is that polarization of bonds and solvation of ions play an enormously important role in determining the reactivity of molecules. In Chapter 37 you will see that radicals are relatively unaffected by solvation and that their reactions follow bond strengths much more closely.



Carboxylic acids do not undergo substitution reactions under basic conditions

Substitution reactions of RCO_2H require a leaving group OH^- . The pK_a of water is about 15, so acids should be about as electrophilic as esters. Esters react well with ammonia to give amides. However, if we try to react carboxylic acids with amines to give amides no substitution occurs: an ammonium salt is formed because the amines themselves are basic and remove the acidic proton from the acid.

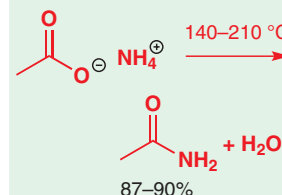


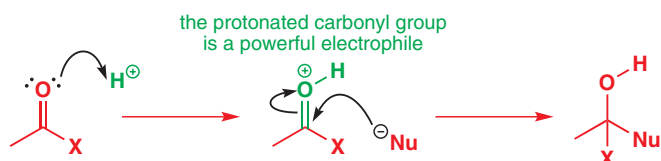
Once the carboxylic acid is deprotonated, substitutions are prevented because (almost) no nucleophile will attack the carboxylate anion. Under neutral conditions, alcohols are just not reactive enough to add to the carboxylic acid but, with *acid* catalysis, esters can be formed from alcohols and carboxylic acids.

Acid catalysts increase the reactivity of a carbonyl group

We saw in Chapter 6 that the lone pairs of a carbonyl group may be protonated by acid. Only strong acids are powerful enough to protonate carbonyl groups: the pK_a of protonated acetone is -7 so, for example, even 1M HCl ($\text{pH } 0$) would protonate only 1 in 10^7 molecules of acetone. However, even proportions as low as this are sufficient to increase the rate of substitution reactions at carbonyl groups enormously because those carbonyl groups that are protonated become extremely powerful electrophiles.

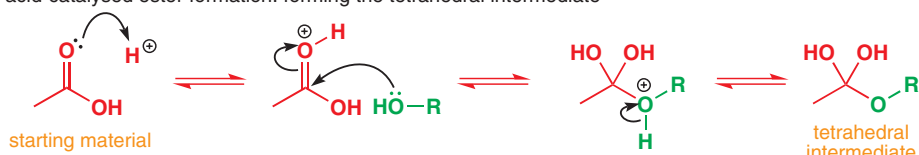
■ In fact, amides *can* be made from carboxylic acids plus amines, but only if the ammonium salt is heated strongly to dehydrate it. This is not usually a good way of making amides!





It is for this reason that alcohols will react with carboxylic acids under acid catalysis. The acid (usually HCl or H₂SO₄) reversibly protonates a small percentage of the carboxylic acid molecules, and the protonated carboxylic acids are extremely susceptible to attack by even a weak nucleophile such as an alcohol. This is the first half of the reaction:

acid-catalysed ester formation: forming the tetrahedral intermediate

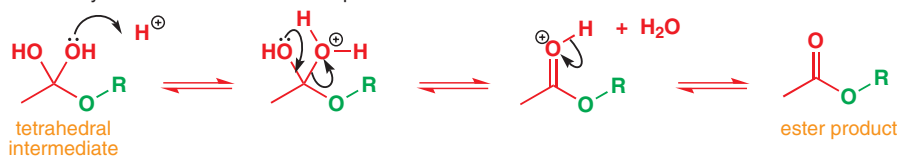


Acid catalysts can make bad leaving groups into good ones

■ Average bond strength C=O
720 kJ mol⁻¹.
Average bond strength C—O
350 kJ mol⁻¹.

This tetrahedral intermediate is unstable because the energy to be gained by re-forming a C=O bond is greater than that used in breaking two C—O bonds. As it stands, none of the leaving groups (R⁻, HO⁻, or RO⁻) is very good. However, help is again at hand in the acid catalyst. It can protonate any of the oxygen atoms reversibly. Again, only a very small proportion of molecules are protonated at any one time but, once the oxygen atom of, say, one of the OH groups is protonated, it becomes a much better leaving group (water instead of HO⁻). Loss of ROH from the tetrahedral intermediate is also possible: this leads back to starting materials—hence the equilibrium arrow in the scheme above. Loss of H₂O is more fruitful, and takes the reaction forwards to the ester product.

acid-catalysed ester formation: decomposition of the tetrahedral intermediate

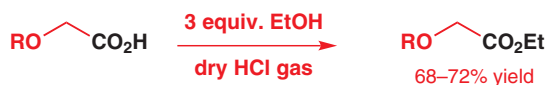


● Acid catalysts catalyse substitution reactions of carboxylic acids.

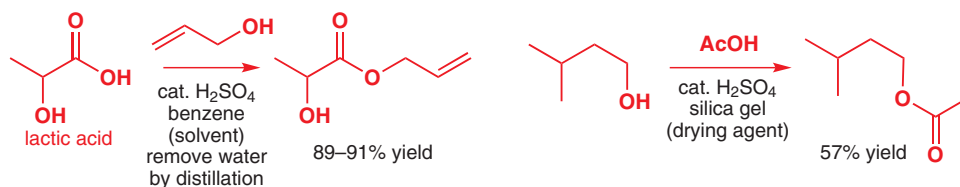
- They make the carbonyl group more electrophilic by protonation at *carbonyl* oxygen.
- They make the leaving group better by protonation there too.

Ester formation is reversible: how to control an equilibrium

Loss of water from the tetrahedral intermediate is reversible too: just as ROH will attack a protonated carboxylic acid, H₂O will attack a protonated ester. In fact, every step in the sequence from carboxylic acid to ester is an equilibrium, and the overall equilibrium constant is about 1. In order for this reaction to be useful, it is therefore necessary to ensure that the equilibrium is pushed towards the ester side by using an excess of alcohol or carboxylic acid (usually the reactions are done in a solution of the alcohol or the carboxylic acid). In this reaction, for example, no water is added and an excess of alcohol is used. Using less than three equivalents of ethanol gave lower yields of ester.



Alternatively, the reaction can be done in the presence of a dehydrating agent (concentrated H_2SO_4 , for example, or silica gel) or the water can be distilled out of the mixture as it forms.



■ Lactic acid must be handled in solution in water. Can you see why, bearing in mind what we have said about the reversibility of ester formation?

● Making esters from alcohols

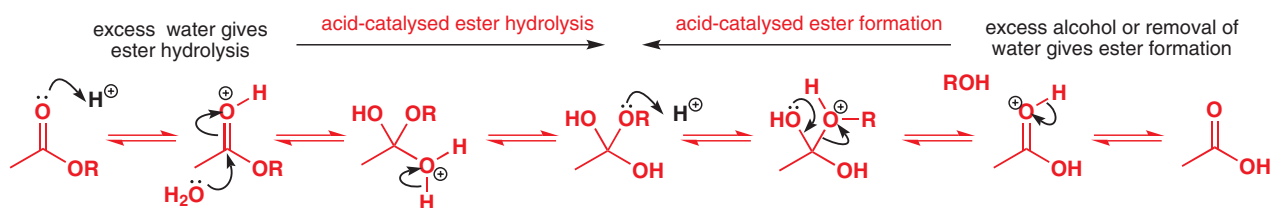
You have now met three ways of making esters from alcohols:

- with acyl chlorides
- with acid anhydrides
- with carboxylic acids.

Try to appreciate that different methods will be appropriate at different times. If you want to make a few milligrams of a complex ester, you are much more likely to work with a reactive acyl chloride or anhydride, using pyridine as a weakly basic catalyst, than to try to distil out a minute quantity of water from a reaction mixture containing a strong acid that may destroy the starting material. On the other hand, if you are a chemist making simple esters (such as those in Chapter 2, p. 31) for the flavouring industry on a scale of many tons, you might prefer the cheaper option of carboxylic acid and a strong acid (e.g. H_2SO_4) in alcohol solution.

Acid-catalysed ester hydrolysis and transesterification

By starting with an ester, an excess of water, and an acid catalyst we can persuade the reverse reaction to occur: formation of the carboxylic acid plus alcohol with consumption of water. Such a reaction is known as a hydrolysis reaction because water is used to break up the ester into carboxylic acid plus alcohol (*lysis*=breaking).

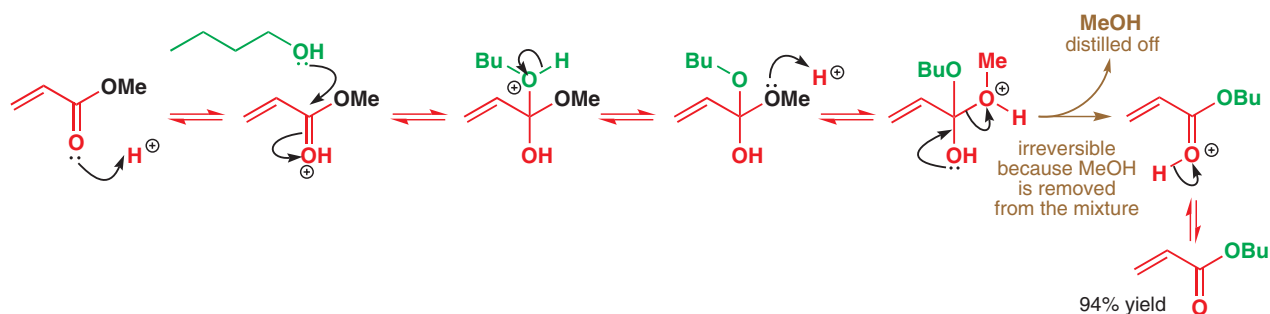


Acid-catalysed ester formation and hydrolysis are the exact reverse of one another: the only way we can control the reaction is by altering concentrations of reagents to drive the reaction the way we want it to go. The same principles can be used to convert an ester of one alcohol into an ester of another, a process known as transesterification. It is possible, for example, to force this equilibrium to the right by distilling methanol (which has a lower boiling point than the other components of the reaction) out of the mixture.

 Interactive mechanism for acid-catalysed ester formation

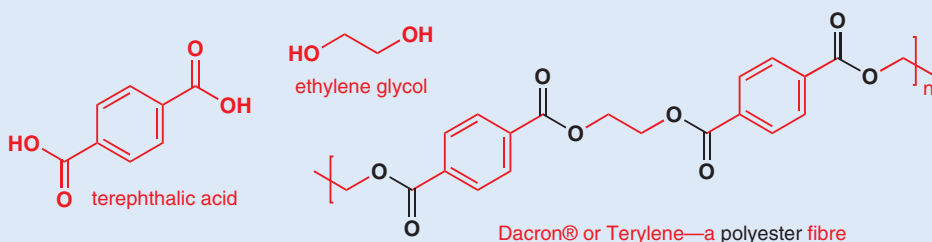


The mechanism for this transesterification simply consists of adding one alcohol (here BuOH) and eliminating the other (here MeOH), both processes being acid-catalysed. Notice how easy it is now to confirm that the reaction is *catalytic* in H^+ .

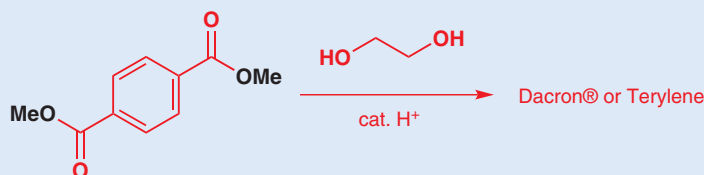


Polyester fibre manufacture

A transesterification reaction is used to make the polyester fibres that are used for textile production. Terylene, or Dacron, for example, is a polyester of the dicarboxylic acid terephthalic acid and the diol ethylene glycol.



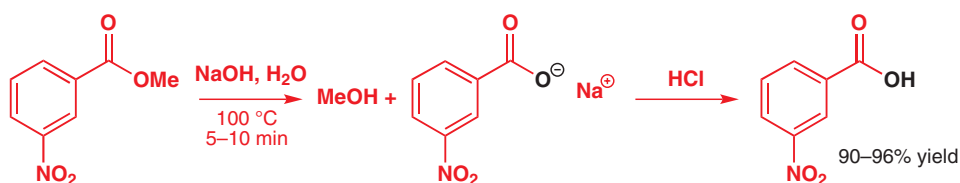
Terylene is actually made by ester exchange: dimethyl terephthalate is heated with ethylene glycol and an acid catalyst, distilling off the methanol as it is formed.



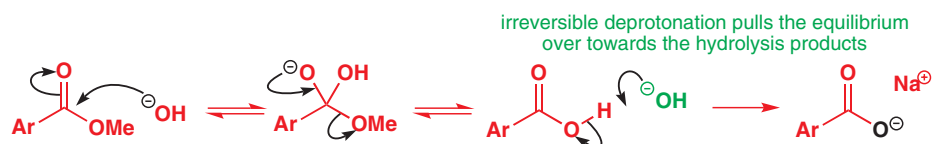
Interactive structure of polyester fibres

Base-catalysed hydrolysis of esters is irreversible

You can't make esters from carboxylic acids and alcohols under basic conditions because the base deprotonates the carboxylic acid (see p. 207). However, you can reverse that reaction and hydrolyse an ester to a carboxylic acid (more accurately, a carboxylate salt) and an alcohol.



This time the ester is, of course, not protonated first as it would be in acid, but the unprotonated ester is a good enough electrophile because OH^- , and not water, is the nucleophile. The tetrahedral intermediate can collapse either way, giving back ester or going forward to acid plus alcohol.

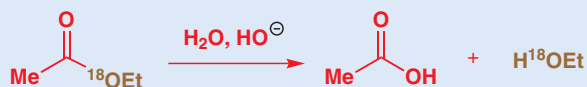


The backward reaction is impossible because the basic conditions straightaway deprotonate the acid to make a carboxylate salt (which, incidentally, consumes the base, making at least one equivalent of base necessary in the reaction). Carboxylate salts do not usually react with nucleophiles, even those a good deal stronger than alcohols.

How do we know this is the mechanism?

Ester hydrolysis is such an important reaction that chemists have spent a lot of time and effort finding out exactly how it works. Many of the experiments that tell us about the mechanism involve oxygen-18 labelling. The starting material is an ester enriched in the heavy oxygen isotope ^{18}O . By knowing where the heavy oxygen atoms start off, and following (by mass spectrometry—Chapter 3) where they end up, the mechanism can be established.

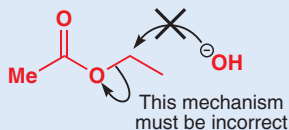
1. An ^{18}O label in the 'ether' oxygen of the ester ends up in the alcohol product.



2. Hydrolysis with $^{18}\text{OH}_2$ gives ^{18}O -labelled carboxylic acid, but no ^{18}O -labelled alcohol.



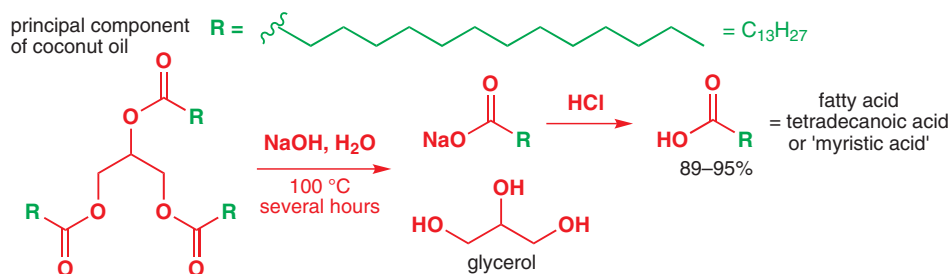
These experiments tell us that a displacement (substitution) has occurred at the carbonyl carbon atom, and rule out the alternative displacement at saturated carbon.



One further labelling experiment showed that a tetrahedral intermediate must be formed: an ester labelled with ^{18}O in its carbonyl oxygen atom passes some of its ^{18}O label to the water. We discussed this on p. 201.

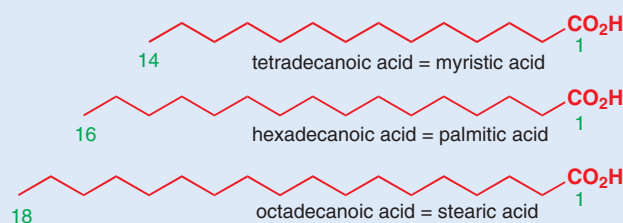
There is more on the mechanism of ester hydrolysis in Chapter 12.

The saturated fatty acid tetradecanoic acid (also known as myristic acid) is manufactured commercially from coconut oil by hydrolysis in base. You may be surprised to learn that coconut oil contains more saturated fat than butter, lard, or beef dripping: much of it is the trimyristate ester of glycerol. Hydrolysis with aqueous sodium hydroxide, followed by reprotonation of the sodium carboxylate salt with acid, gives myristic acid. Notice how much longer it takes to hydrolyse this branched ester than it did to hydrolyse a methyl ester (p. 210).



Saponification

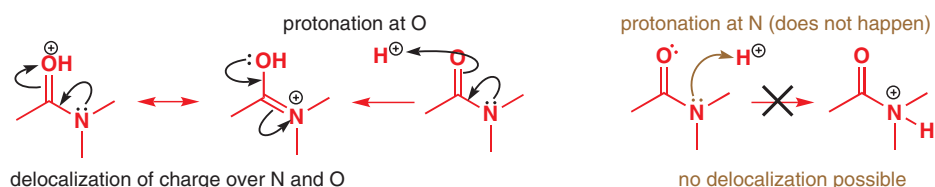
The alkaline hydrolysis of esters to give carboxylate salts is known as saponification because it is the process used to make soap. Traditionally, beef tallow (the tristearate ester of glycerol—stearic acid is octadecanoic acid, $C_{17}H_{35}CO_2H$) was hydrolysed with sodium hydroxide to give sodium stearate, $C_{17}H_{35}CO_2Na$, the principal component of soap. Finer soaps are made from palm oil and contain a higher proportion of sodium palmitate, $C_{15}H_{31}CO_2Na$. Hydrolysis with KOH gives potassium carboxylates, which are used in liquid soaps. Soaps like these owe their detergent properties to the combination of polar (carboxylate group) and non-polar (long alkyl chain) properties.



Amides can be hydrolysed under acidic or basic conditions too

In order to hydrolyse amides, the least reactive of the carboxylic acid derivatives, we have a choice: we can persuade the amine leaving group to leave by protonating it, or we can use brute force and forcibly eject it with concentrated hydroxide solution.

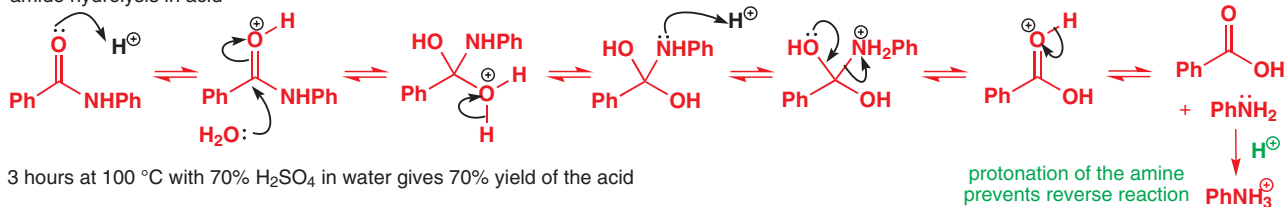
Amides are very unreactive as electrophiles, but they are also rather more basic than most carboxylic acid derivatives: a typical protonated amide has a pK_a of -1 ; most other carbonyl compounds are much less basic. You might therefore imagine that the protonation of an amide would take place on nitrogen—after all, *amine* nitrogen atoms are readily protonated. And, indeed, the reason for the basicity of amides is the nitrogen atom's delocalized lone pair, making the carbonyl group unusually electron rich. But amides are always protonated on the oxygen atom of the carbonyl group, never the nitrogen, because protonation at nitrogen would disrupt the delocalized system that makes amides so stable. Protonation at oxygen gives a delocalized cation (Chapter 8).



Protonation of the carbonyl group by acid makes the carbonyl group electrophilic enough for attack by water, giving a neutral tetrahedral intermediate. The amine nitrogen atom in the tetrahedral intermediate is much more basic than the oxygen atoms, so now *it* gets protonated, and the RNH_2 group becomes really quite a good leaving group. Once it has left, it will immediately be protonated again, and therefore become completely non-nucleophilic. The conditions are very vigorous—70% sulfuric acid for 3 hours at $100^\circ C$.

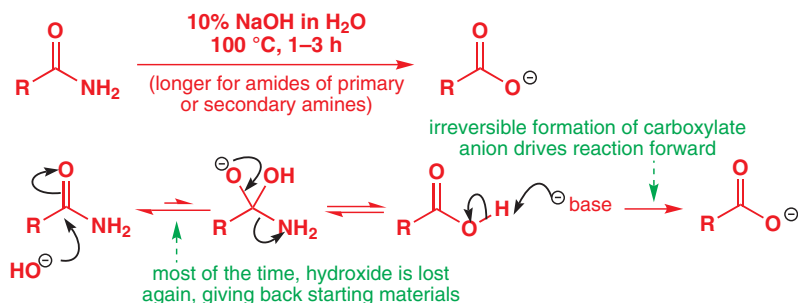
■ Notice that this means that one equivalent of acid is used up in this reaction—the acid is not solely a catalyst.

amide hydrolysis in acid

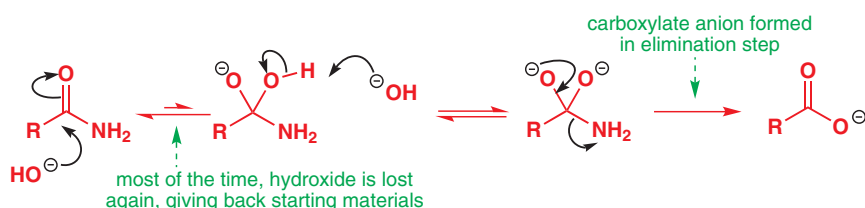


Hydrolysis of amides in base requires similarly vigorous conditions. Hot solutions of hydroxide are sufficiently powerful nucleophiles to attack an amide carbonyl group, although even when the tetrahedral intermediate has formed NH_2^- ($\text{p}K_{\text{a}}$ of the ammonium ion 35) has only a slight chance of leaving when HO^- ($\text{p}K_{\text{a}}$ of water 15) is an alternative. Nonetheless, at high temperatures amides are slowly hydrolysed by concentrated base since one product is the carboxylate salt and this does not react with nucleophiles. The 'base' for the irreversible step might be hydroxide or NH_2^- .

amide hydrolysis in base



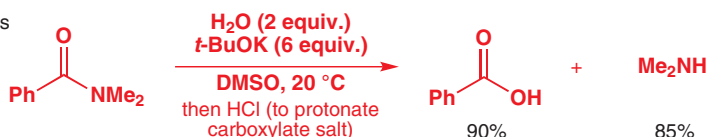
Secondary and tertiary amides hydrolyse much more slowly under these conditions. With all these amides a second mechanism kicks in if the hydroxide concentration is large enough. More hydroxide deprotonates the tetrahedral anion to give a dianion that must lose NH_2^- as the only alternative is O^{2-} . This leaving group deprotonates water so the second molecule of hydroxide ion is simply a catalyst.



■ You've not seen the option of O^{2-} as a leaving group before but this is what you would need if you want to break the bond to O^- . Asking O^{2-} to be a leaving group is like asking HO^- to be an acid.

A similar mechanism is successful with only a little water and plenty of strong base. Then even tertiary amides can be hydrolysed at room temperature. Potassium *tert*-butoxide is a strong enough base ($\text{p}K_{\text{a}}$ of *t*-BuOH about 18) to deprotonate the tetrahedral intermediate.

hydrolysis of amides using *t*-BuOK



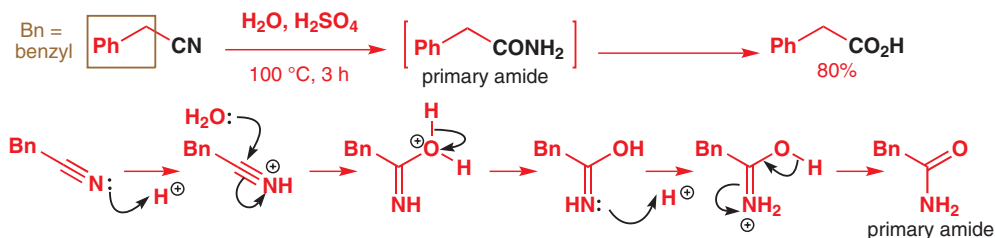
Hydrolysing nitriles: how to make the almond extract, mandelic acid

Closely related to the amides are nitriles. You can view them as primary amides that have lost one molecule of water and, indeed, they can be made by dehydrating primary amides.



They can be hydrolysed just like amides too. Addition of water to the protonated nitrile gives a primary amide, and hydrolysis of this amide gives carboxylic acid plus ammonia.

■ Don't be put off by the number of steps in this mechanism—look carefully and you will see that most of them are simple proton transfers. The only step that isn't a proton transfer is the addition of water.



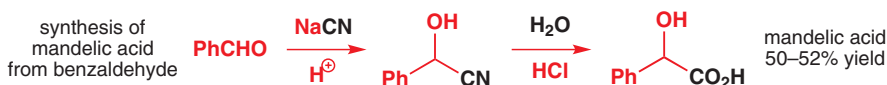
You met a way of making nitriles—from HCN (or NaCN + HCl) plus aldehydes—in Chapter 6: the hydroxynitrile products are known as **cyanohydrins**. With this in mind, you should be able to suggest a way of making mandelic acid, an extract of almonds, from benzaldehyde.

reminder from Chapter 6:



This is how some chemists did it.

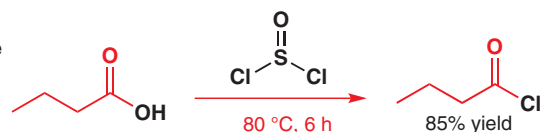
■ You have just designed your first total synthesis of a natural product. We return to designing syntheses much later in this book, in Chapter 28.



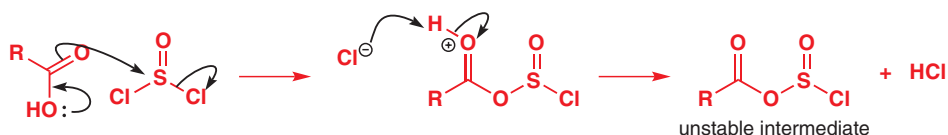
Acid chlorides can be made from carboxylic acids using SOCl_2 or PCl_5

We have looked at a whole series of interconversions between carboxylic acid derivatives and, after this next section, we shall summarize what you need to understand. We said that it is always easy to move down the series of acid derivatives we listed early in the chapter, and so far that is all we have done. But some reactions of carboxylic acids also enable us to move upwards in the series. What we need is a reagent that changes the bad leaving group HO^- into a good leaving group. Strong acid does this by protonating the OH^- , allowing it to leave as H_2O . In this section we look at two more reagents, SOCl_2 and PCl_5 , which convert the OH group of a carboxylic acid and also turn it into a good leaving group. Thionyl chloride, SOCl_2 , reacts with carboxylic acids to make acyl chlorides.

acid chlorides can be made from carboxylic acids with thionyl chloride

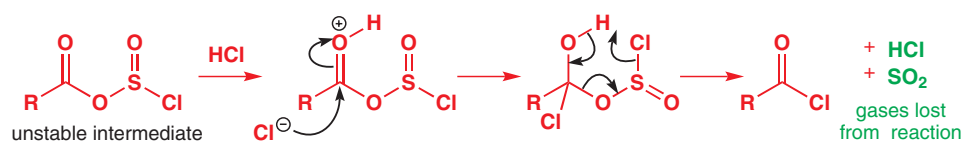


This volatile liquid with a choking smell is electrophilic at the sulfur atom (as you might expect with two chlorine atoms and an oxygen atom attached) and is attacked by carboxylic acids to give an unstable, and highly electrophilic, intermediate.



Reprotonation of the unstable intermediate (by the HCl just produced, i.e. reversal of the last step above) gives an electrophile powerful enough to react even with the weak nucleophile Cl^- (HCl is a strong acid, so Cl^- is a poor nucleophile). The tetrahedral intermediate collapses to the acyl chloride, sulfur dioxide, and hydrogen chloride. This step is irreversible because SO_2 and HCl are gases that are lost from the reaction mixture.

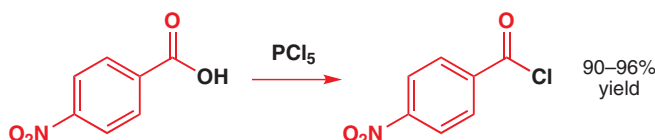
■ Note that it is the more nucleophilic carbonyl oxygen which actually attacks S. If you follow the fate of the two oxygens right through the mechanism you will see which fact it is the oxygen that starts off in the $\text{C}=\text{O}$ group which is replaced by Cl. You may also be surprised to see the way we substituted at $\text{S}=\text{O}$ without forming a 'tetrahedral intermediate'. Well, this trivalent sulfur atom is already tetrahedral (it still has one lone pair), and substitution can go by a direct substitution at sulfur.



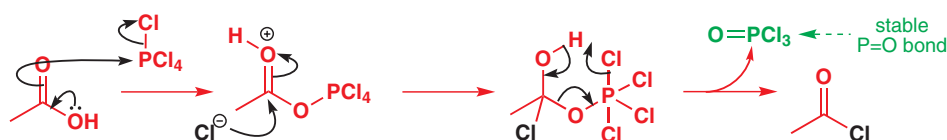
Interactive mechanism for acid chloride formation with SOCl_2

Although HCl is involved in this reaction, it cannot be used as the sole reagent for making acid chlorides. It is necessary to have a sulfur or phosphorus compound to remove the oxygen. An alternative reagent for converting RCO_2H into RCOCl is phosphorus pentachloride, PCl_5 . The mechanism is similar—try writing it out before looking at the scheme below.

acid chlorides can be made from carboxylic acids with phosphorus pentachloride



The mechanism is closely related to the previous one, except that the formation of a very stable $\text{P}=\text{O}$ bond is the vital factor rather than the loss of two gaseous reagents.

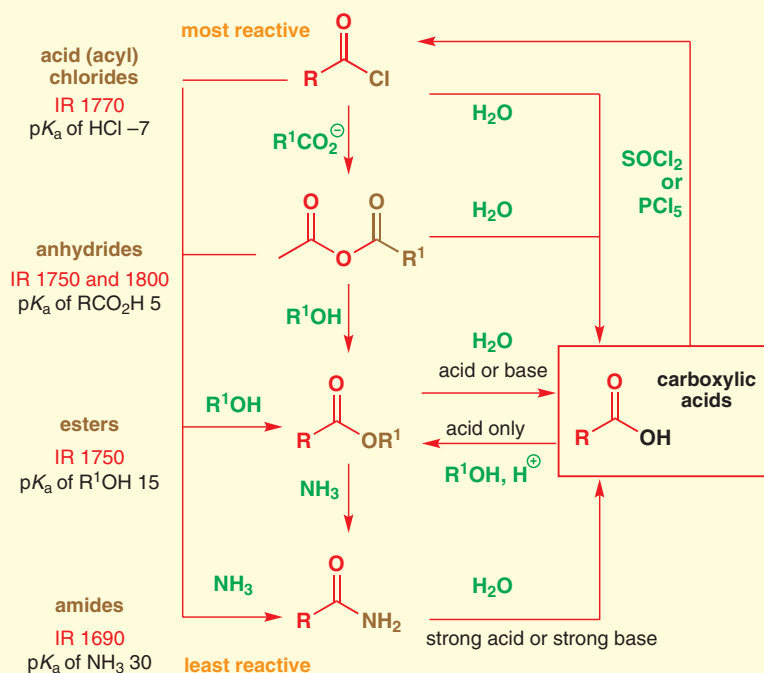


Interactive mechanism for acid chloride formation with PCl_5

These conversions of acids into acid chlorides complete all the methods we need to convert acids into any acid derivatives. You can convert acids directly to esters and now to acid chlorides, the most reactive of acid derivatives, and can make any other derivative from them. The chart below adds reaction conditions, relevant pK_a s, and infrared stretching frequencies to the reactivity order we met earlier.

■ We will explore the link between infrared stretching frequency and reactivity in Chapter 18.

● Interconversion of carboxylic acid derivatives



All these acid derivatives can, of course, be hydrolysed to the acid itself with water alone or with various levels of acid or base catalysis depending on the reactivity of the derivative.

To climb the reactivity order therefore, the simplest method is to hydrolyse to the acid and convert the acid into the acid chloride. You are now at the top of the reactivity order and can go down to whatever level you require.

Making other compounds by substitution reactions of acid derivatives

■ Five 'oxidation levels'—(1) hydrocarbon, (2) alcohol, (3) aldehyde and ketone, (4) carboxylic acid, and (5) CO_2 —were defined in Chapter 2.

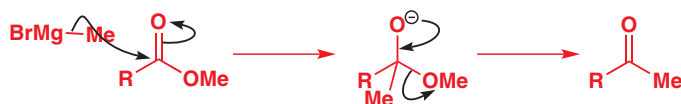
We've talked at length about the interconversions of acid derivatives, explaining the mechanism of attack of nucleophiles such as ROH , H_2O , and NH_3 on acyl chlorides, acid anhydrides, esters, acids, and amines, with or without acid or base present. We shall now go on to talk about substitution reactions of acid derivatives that take us out of this closed company of compounds and allow us to make compounds containing functional groups at other oxidation levels, such as ketones and alcohols.

Making ketones from esters: the problem

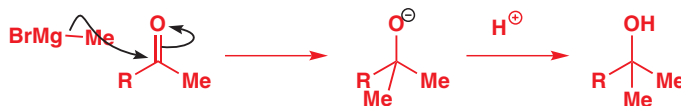
Substitution of the OR group of an ester by an R group would give us a ketone. You might therefore think that reaction of an ester with an organolithium or Grignard reagent would be a good way of making ketones. However, if we try the reaction, something else happens, as you saw at the start of this chapter.



Two molecules of Grignard have been incorporated and we get an alcohol! If we look at the mechanism we can understand why this should be so. First, as you would expect, the nucleophilic Grignard reagent attacks the carbonyl group to give a tetrahedral intermediate. The only reasonable leaving group is RO^- , so it leaves to give us the ketone we set out to make.



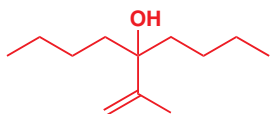
Now, the next molecule of Grignard reagent has a choice. It can react with either the ester starting material or the newly formed ketone. Ketones are more electrophilic than esters so the Grignard reagent prefers to react with the ketone in the manner you saw in Chapter 9. A stable alkoxide anion is formed, which gives the tertiary alcohol on acid work-up.

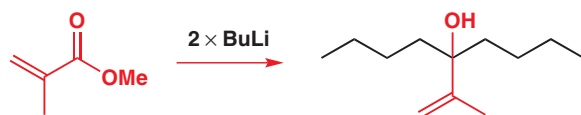


Making alcohols instead of ketones

In other words, the problem here lies in the fact that the ketone product is more reactive than the ester starting material. We shall meet more examples of this general problem later (in Chapter 23, for example): in the next section we shall look at ways of overcoming it. Meanwhile, why not see it as a useful reaction? This compound, for example, was needed by some chemists in the course of research into explosives.

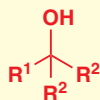
It is a tertiary alcohol with the hydroxyl group flanked by two identical R (= butyl) groups. The chemists who wanted to make the compound knew that an ester would react twice with the same organolithium reagent, so they made it from this unsaturated ester (known as methyl methacrylate) and butyllithium.





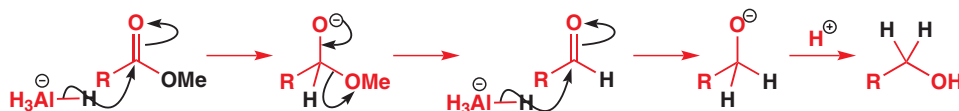
• Tertiary alcohol synthesis

Tertiary alcohols with two identical R^2 groups can be made from ester R^1CO_2R plus two equivalents of organolithium R^2Li or Grignard reagent R^2MgBr .



This reaction works in reduction too if we use lithium aluminium hydride, $LiAlH_4$. This is a powerful reducing agent that readily attacks the carbonyl group of an ester. Again, collapse of the tetrahedral intermediate gives a compound, this time an aldehyde, which is more reactive than the ester starting material, so a second reaction takes place and the ester is converted (reduced) into an alcohol. Sodium borohydride, often used for the reduction of ketones, does not usually reduce esters.

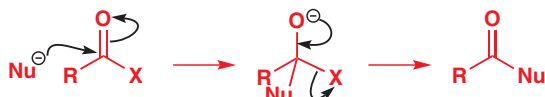
reduction of esters by $LiAlH_4$



This is an extremely important reaction, and one of the best ways of making alcohols from esters. Stopping the reaction at the aldehyde stage is more difficult: we shall discuss this in Chapter 23.

A bit of shorthand

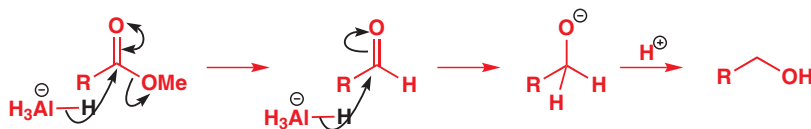
Before we go any further, we should introduce to you a little bit of chemical shorthand that makes writing many mechanisms easier. As you now appreciate, all substitution reactions at a carbonyl group go via a tetrahedral intermediate.



A convenient way to save writing a step is to show the formation and collapse of the tetrahedral intermediate in the same structure, by using a double-headed arrow, as in the diagrams below. Now, this is a useful shorthand, but it is not a substitute for understanding the true mechanism. Certainly, you must never ever write the reaction as a single step not involving the carbonyl group.



Here's the 'shorthand' at work in the $LiAlH_4$ reduction you have just met.



Making ketones from esters: the solution

We diagnosed the problem with our intended reaction as one of reactivity: the product ketone is more reactive than the starting ester. To get round this problem we need to do one of two things:

1. make the starting material more reactive *or*
2. make the product less reactive.

Making the starting materials more reactive

A more reactive starting material would be an acyl chloride: how about reacting one of these with a Grignard reagent? This approach can work—for example this reaction is successful.



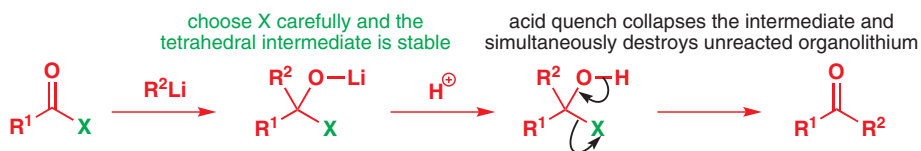
Often, better results are obtained by transmetallating (see Chapter 9) the Grignard reagent, or the organolithium, with copper salts. Organocopper reagents are too unreactive to add to the product ketones, but they react well with the acyl chloride. Consider this reaction, for example: the product was needed for a synthesis of the antibiotic septamycin.



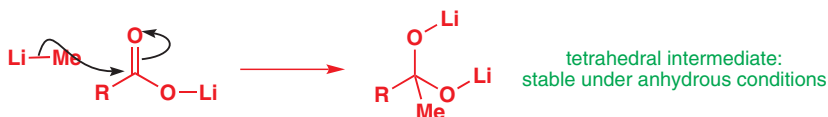
■ Notice how this reaction illustrates the difference in reactivity between an acyl chloride functional group and an ester functional group.

Making the products less reactive

This alternative solution is often better. With the right starting material, the tetrahedral intermediate can become stable enough not to collapse to a ketone during the reaction; it therefore remains completely unreactive towards nucleophiles. The ketone is formed only when the reaction is finally quenched with acid but the nucleophile is also destroyed by the acid and none is left for further addition.

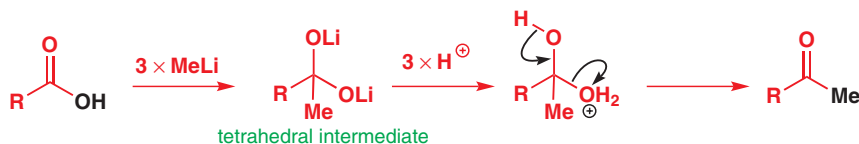


We can illustrate this concept with a reaction of an unlikely looking electrophile, a lithium carboxylate. Towards the beginning of the chapter we said that carboxylic acids were bad electrophiles and that carboxylate salts were even worse. Well, that is true, but with a sufficiently powerful nucleophile (an organolithium) it is just possible to get addition to the carbonyl group of a lithium carboxylate.

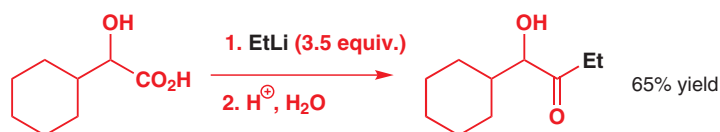


We could say that the affinity of lithium for oxygen means that the Li-O bond has considerable covalent character, making the CO_2Li less of a true anion. And the intermediate after addition of MeLi is probably best represented as a covalent compound too. Anyway, the

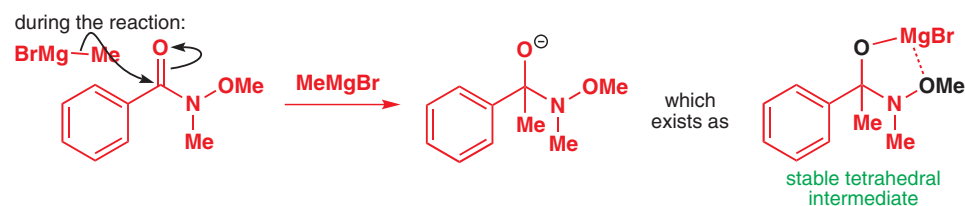
product of this addition is a dianion of the sort that we met during one of the mechanisms of base-catalysed amide hydrolysis. But in this case there is no possible leaving group, so there the dianion sits. Only at the end of the reaction, when water is added, are the oxygen atoms protonated to give a hydrated ketone, which collapses immediately (remember Chapter 6) to give the ketone that we wanted. The water quench also destroys any remaining organolithium, so the ketone is safe from further attack.



This method has been used to make some ketones that are important starting materials for making cyclic natural products known as macrolides.

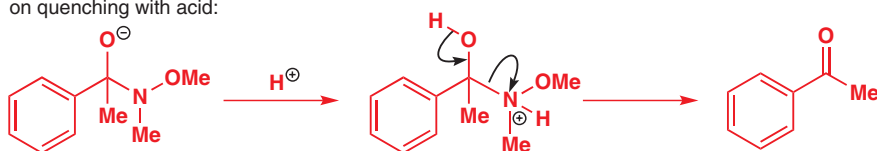


Another good set of starting materials that lead to non-collapsible tetrahedral intermediates is known as the **Weinreb amides**, after their inventor, S. M. Weinreb. Addition of organolithium or organomagnesium reagents to *N*-methoxy-*N*-methyl amides gives the tetrahedral intermediate shown, stabilized by *chelation* of the magnesium atom by the two oxygen atoms. Chelation means the coordination of more than one electron-donating atom in a molecule to a single metal atom.

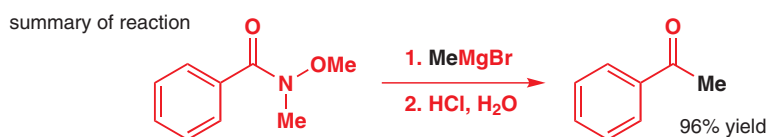


This intermediate collapses to give a ketone only when acid is added at the end of the reaction.

on quenching with acid:



The mechanism looks complicated but the reaction is easy to do:

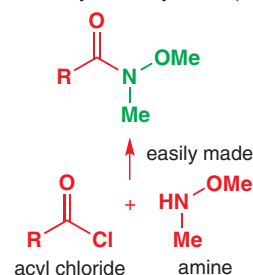


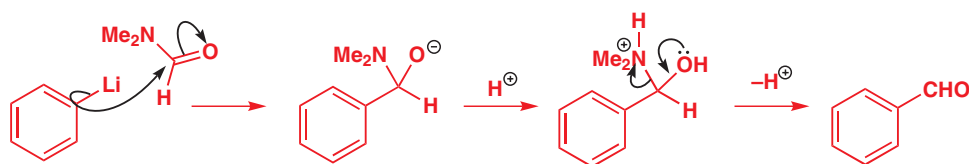
This strategy even works for making aldehydes, if the starting material is dimethylformamide (DMF, Me_2NCHO). This is an extremely useful way of adding electrophilic CHO groups to organometallic nucleophiles. Once again, the tetrahedral intermediate is stable until acid is added at the end of the reaction and the protonated tetrahedral intermediate collapses.

■ Notice that three equivalents of organolithium are needed in this reaction: one to deprotonate the acid, one to deprotonate the hydroxyl group, and one to react with the lithium carboxylate. These chemists added a further 0.5 for good measure.

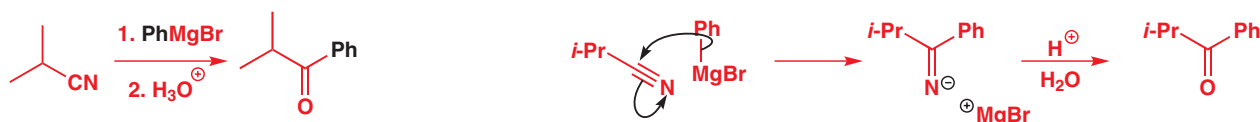
■ The word chelation derives from *chele*, the Greek for 'claw'.

a Weinreb amide (an *N*-methoxy-*N*-methyl amide)



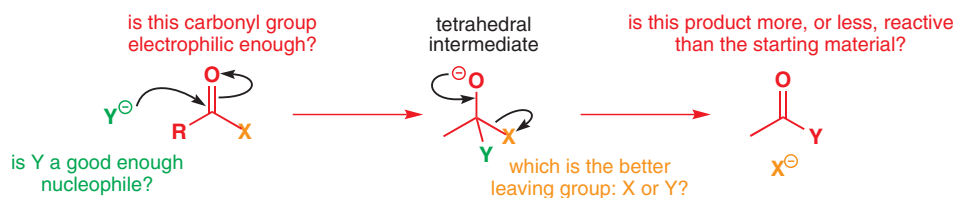


A final alternative is to use a nitrile instead of an ester. The intermediate is the anion of an imine (see Chapter 12 for more about imines), which is not electrophilic at all—in fact, it's quite nucleophilic, but there are no electrophiles for it to react with until the reaction is quenched with acid. It gets protonated and hydrolyses (we'll discuss this in the next chapter) to the ketone.



To summarize...

To finish, we should just remind you of what to think about when you consider a nucleophilic substitution at a carbonyl group.



And to conclude...

In this chapter you have been introduced to some important reactions—you can consider them to be a series of facts if you wish, but it is better to see them as the logical outcome of a few simple mechanistic steps. Relate what you have seen to what you gathered from Chapters 6 and 9, when we first started looking at carbonyl groups. All we did in this chapter was to build some subsequent transformations on to the simplest organic reaction, addition to a carbonyl group. You should have noticed that the reactions of all acid derivatives are related and are very easily explained by writing out proper mechanisms, taking into account the presence of acid or base. In the next two chapters we shall see more of these acid- and base-catalysed reactions of carbonyl groups. Try to view them as closely related to the ones in this chapter—the same principles apply to their mechanisms.

Further reading

Section 2, 'Nucleophilic substitution to the carbonyl group' in S. Warren, *Chemistry of the Carbonyl Group*, Wiley, Chichester, 1974.

The dehydration of amides to give nitriles is described in Vogel, p. 716.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>